MODEL LAB · 2026-10-13

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

28,643
held-out plate appearances

3.05%
held-out HR rate

0.5958
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-01, scored on the 28,643 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5777	1.60×	0.02950	0.13551	0.00489
KNN comps Prior is the pooled rate of the nearest comparables.	0.5958	1.48×	0.02946	0.13462	0.00484
SVR Support-vector regression on the same feature matrix.	0.5884	1.42×	0.02950	0.13532	0.00650
Random forest Forest regression; a learned similarity metric.	0.5894	1.44×	0.02949	0.13501	0.00627
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5958	1.48×	0.02946	0.13462	0.00484
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5957	1.47×	0.02946	0.13462	0.00483
Linear regression Ordinary least squares on the same features.	0.5900	1.42×	0.02948	0.13505	0.00535
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5906	1.49×	0.02950	0.13515	0.00657
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5896	1.48×	0.02949	0.13505	0.00660
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5904	1.41×	0.02950	0.13513	0.00653

BLEND WEIGHTS (FOREST VARIANT)

KNN **0.90** SVR **0.00** RF **0.10**

SHRINKAGE K PER PRIOR

league **249** knn **249** svr **249** rf **249** core **249** ensemble **249** linear **249** logistic **249** nn **249** xgb **249**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,116 PA, actual HR rate 2.37%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5823	1.48×	0.02319	0.11242	0.01187
KNN comps Prior is the pooled rate of the nearest comparables.	0.5746	1.08×	0.02323	0.11256	0.01296
SVR Support-vector regression on the same feature matrix.	0.5684	1.21×	0.02342	0.11561	0.01853
Random forest Forest regression; a learned similarity metric.	0.5740	1.21×	0.02334	0.11344	0.01619
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5746	1.08×	0.02323	0.11256	0.01296
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5744	1.08×	0.02324	0.11250	0.01259
Linear regression Ordinary least squares on the same features.	0.5766	0.94×	0.02333	0.11448	0.01600
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5846	1.08×	0.02332	0.11370	0.01486
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5721	1.21×	0.02329	0.11362	0.01393
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5843	1.21×	0.02337	0.11380	0.01667

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Jo Adell 16.0%, Junior Caminero 15.8%, Jonathan Aranda 12.9%, Chase DeLauter 12.2%, Yandy Díaz 11.8%, Victor Mesa Jr. 11.5%
Flat league	6/6	Jo Adell 15.2%, Junior Caminero 14.9%, Chase DeLauter 12.3%, Yandy Díaz 12.2%, Jonathan Aranda 12.1%, Victor Mesa Jr. 11.4%
KNN comps	6/6	Jo Adell 15.7%, Junior Caminero 15.6%, Jonathan Aranda 13.1%, Chase DeLauter 12.2%, Victor Mesa Jr. 11.8%, Yandy Díaz 11.8%
SVR	6/6	Junior Caminero 17.2%, Jo Adell 16.2%, Jonathan Aranda 12.6%, Victor Mesa Jr. 12.3%, Yandy Díaz 12.2%, Chase DeLauter 12.1%
Random forest	6/6	Junior Caminero 16.1%, Jo Adell 15.8%, Jonathan Aranda 12.7%, Chase DeLauter 12.2%, Yandy Díaz 12.1%, Victor Mesa Jr. 11.9%
Core (KNN+SVR)	6/6	Jo Adell 15.7%, Junior Caminero 15.6%, Jonathan Aranda 13.1%, Chase DeLauter 12.2%, Victor Mesa Jr. 11.8%, Yandy Díaz 11.8%
Forest blend	6/6	Jo Adell 15.7%, Junior Caminero 15.6%, Jonathan Aranda 13.0%, Chase DeLauter 12.2%, Yandy Díaz 11.8%, Victor Mesa Jr. 11.8%
Linear regression	6/6	Jo Adell 15.9%, Junior Caminero 15.7%, Jonathan Aranda 12.9%, Yandy Díaz 12.0%, Chase DeLauter 12.0%, Victor Mesa Jr. 11.9%
Logistic regression	6/6	Jo Adell 16.2%, Junior Caminero 16.1%, Jonathan Aranda 12.7%, Chase DeLauter 11.9%, Victor Mesa Jr. 11.9%, Yandy Díaz 11.8%
Neural net (torch)	6/6	Jo Adell 16.3%, Junior Caminero 16.2%, Jonathan Aranda 12.7%, Victor Mesa Jr. 12.0%, Chase DeLauter 11.9%, Yandy Díaz 11.9%
XGBoost	6/6	Junior Caminero 16.6%, Jo Adell 16.0%, Jonathan Aranda 12.9%, Yandy Díaz 12.3%, Chase DeLauter 12.0%, Victor Mesa Jr. 11.7%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	**Jo Adell** CLE slate	**16.0%**	15.2%	15.7%	16.2%	15.8%	15.7%	15.7%	15.9%	16.2%	
2	**Junior Caminero** TB slate	**15.8%**	14.9%	15.6%	17.2%	16.1%	15.6%	15.6%	15.7%	16.1%	
3	**Jonathan Aranda** TB	**12.9%**	12.1%	13.1%	12.6%	12.7%	13.1%	13.0%	12.9%	12.7%	
4	**Chase DeLauter** CLE	**12.2%**	12.3%	12.2%	12.1%	12.2%	12.2%	12.2%	12.0%	11.9%	
5	**Yandy Díaz** TB	**11.8%**	12.2%	11.8%	12.2%	12.1%	11.8%	11.8%	12.0%	11.8%	
6	**Victor Mesa Jr.** TB	**11.5%**	11.4%	11.8%	12.3%	11.9%	11.8%	11.8%	11.9%	11.9%	
7	**Nathaniel Lowe** CLE	**11.5%**	11.3%	11.6%	11.4%	11.7%	11.6%	11.6%	11.5%	11.4%	
8	**Travis Bazzana** CLE	**11.2%**	11.3%	11.1%	10.9%	11.3%	11.1%	11.2%	11.1%	10.8%	
9	**José Ramírez** CLE	**10.8%**	10.8%	10.7%	10.8%	10.7%	10.7%	10.7%	10.9%	11.2%	
10	**Petey Halpin** CLE	**9.2%**	10.3%	9.4%	9.8%	9.5%	9.4%	9.4%	10.3%	9.8%	
11	**Austin Hedges** CLE	**8.6%**	10.4%	9.1%	9.1%	9.2%	9.1%	9.1%	9.4%	9.0%	
12	**Jonny DeLuca** TB	**8.4%**	9.2%	8.3%	8.4%	7.7%	8.3%	8.2%	8.2%	8.5%	
13	**Brayan Rocchio** CLE	**7.3%**	7.8%	7.0%	7.3%	6.9%	7.0%	7.0%	7.0%	7.0%	
14	**Steven Kwan** CLE	**5.4%**	8.7%	6.2%	3.8%	4.6%	6.2%	6.0%	3.8%	5.4%	
15	**Ben Williamson** TB	**5.2%**	8.3%	5.5%	5.1%	5.7%	5.5%	5.5%	3.6%	5.3%	
16	**Taylor Walls** TB	**4.9%**	6.8%	4.9%	4.3%	4.3%	4.9%	4.9%	5.1%	5.1%	
17	**Liam Hicks** TB	**4.7%**	5.7%	4.8%	5.5%	4.9%	4.8%	4.8%	5.1%	5.0%	
18	**Chandler Simpson** TB	**2.0%**	4.6%	2.8%	1.1%	1.3%	2.8%	2.7%	1.1%	2.3%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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